

Applicant: LUO Xiaoyu

Program: MSc Information Technology, The Hong Kong Polytechnic University

PERSONAL STATEMENT

My initial understanding of information technology stemmed from its ability to provide me with a powerful means of information acquisition. Starting from independently searching for resources on the internet in the early days, to personally assembling the first computer and exploring the principles of hardware coordination and operating systems, I underwent a transformation from a user to an explorer. Nowadays, witnessing the breakthrough developments in technologies like artificial intelligence has convinced me that I should be a driver of this transformation. I am eager to systematically build my knowledge system in the MSc in Information Technology program at The Hong Kong Polytechnic University, aiming at achieving a transformation from a developer to an innovator, and honing the ability to build a truly inclusive and trustworthy digital future.

My undergraduate learning, through systematic course learning and project practice, has provided me with a solid foundation for entering the field of information technology. In courses such as *Introduction to Software Engineering* and *Algorithm Design and Analysis*, I gradually developed a structured understanding of system architecture and program design; while in practical courses like *Python Programming Foundation* and *Web Application Development*, I continuously transformed theory into executable code, enhancing my development and debugging capabilities. During a team project to develop a traffic query system, I voluntarily took on the front-end development task. By referring to the official documentation and open-source projects of Vue framework, I quickly mastered the component-based development method and independently completed the interface design and function implementation. This experience made me deeply realize that in the rapidly evolving technology field, continuous learning and the ability to integrate knowledge are of utmost importance. Therefore, I established my personal GitHub knowledge base, systematically organizing the project code and technical notes, and gradually forming a reusable and iterative development system. This not only helped me reuse and improve existing experience in subsequent projects but also allowed me to gradually establish my own development system and learning methods through continuous accumulation and reflection. Now, I am well aware that to achieve the transition from implementing functions to developing systems in the field of information technology, it is necessary to further solidify the knowledge in core areas such as algorithms, artificial intelligence, and distributed systems.

Realizing this, I have been keeping with the passion for investigating information technology in a great scenario. During the one-month server operation internship, I directly encountered the gap between theoretical plans and engineering reality. My main task was to write data migration verification scripts. Initially, the scheme based on configuration files failed because the backup files were dynamically updated. Faced with this situation, I adjusted the scheme to be based on querying the server's purchase time through API and determining whether the data migration was successful. After many attempts and adjustments, I successfully solved the problem. This experience made me deeply understand that an excellent technical solution must strike a balance between theoretical correctness, security constraints, and execution efficiency. Additionally, in an automatic backup task of the bastion host, I mistakenly deleted the configuration files due to excessive permissions. Under the guidance of senior colleagues, I learned to locate anomalies by inserting logs in segments and gradually established a systematic troubleshooting mindset. This experience not only strengthened my understanding of the importance of permission management and production environment security but also helped me develop the necessary rigor and calmness when facing complex system problems. Since then, I have come to understand that modern information technology is no longer just a breakthrough in a single technological aspect, but rather a system-level design, management and collaboration. This is precisely the core that I am eager to delve into in the MSc in Information Technology program at the Hong Kong Polytechnic University.

This MSc Information Technology program, with its emphasis on core areas such as software engineering, information system security and distributed computing, perfectly aligns with my current cognitive and developmental needs. Its curriculum integrates computer science principles with the latest technology applications, and particularly focuses on

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developing students' ability to solve complex IT problems, exactly the core competency I want to strengthen in my future career. In the practice of MySQL database development, I have accumulated experience in managing relational databases. This experience has taught me that to build a reliable data solution, systematic theoretical support is necessary. The course Data Structures and Database Systems course, resonating with this experience, will enable me to gain a deep understanding of the ACID transaction mechanism, the scalability trade-offs of MySQL, and the principles of distributed transactions. This knowledge will enable me to have the ability to participate in designing robust data layer solutions that can support high-concurrency business in the future, rather than merely writing scripts to operate them. In addition, I deem the course Software Engineering and Development is in perfect accordance with my current demands. During my internship, I witnessed the failure of the plan in a complex system, which made me deeply realize my lack of systematic engineering skills. Through this course, I will learn how to apply agile development and DevOps concepts to standardize the project lifecycle and avoid any operational accidents caused by unauthorized access again. In summary, I believe that this program at PolyU will provide me with a systematic set of theories and tools, enabling me not only to solve a technical problem in the future but also to prevent such issues from the beginning of the design, and to build a truly reliable and scalable IT system.

After completing my master's degree, I hope to enter the information technology or software development field, working on system architecture design or intelligent application development. Within 3–5 years, I aim to grow into a technical engineer or R&D lead who can independently manage large-scale projects, integrate cutting-edge technologies, and optimize system performance.

Last but not least, your program will not only allow me to systematically improve my comprehensive abilities in computer science and information technology but also develop the breadth and depth of my thinking in engineering practice and technological innovation. I believe this will provide solid academic support and a practical platform for me to achieve my career goals, enabling me to constantly surpass myself in the rapidly developing information technology field and grow into a truly innovative and influential technical professional.

LUO Xiaoyu

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EDUCATION

Aberdeen Institute of Data Science and Artificial Intelligence, South China Normal University

Bachelor of Engineering in Software Engineering, South China Normal University

Bachelor of Science with Honours in Computing Science, University of Aberdeen

09/2022-07/2026

- GPA: 3.27/5.0 (82.7/100)
- Courses: *Computer Architecture, Java Programming, Web Application Development, Software Testing and Quality, Artificial Intelligence Foundation, Data Mining and Visualization, etc*

INTERNSHIPS

Intern, Beijing Yanling Network Wei Intelligent Technology Co., Ltd.

07/2025-09/2025

- Led in composing project planning based on referring to the existing research findings, and collected analysis data to check the feasibility of the codes edited
- Composed a script to extract vulnerability information and analyzed its preciseness and efficiency, aiming at optimizing the script's performance and predicting the overall cost and performance consumption ratio
- Applied LLM to classify and deliver a vulnerability detection to the Web applications and mini-programs, and utilized the taint analysis technique to trace the propagation path of tainted data within the program

Intern, Guangzhou Netlaw Internet Technology Co., Ltd.

10/2024-12/2024

Technique Engagement

- Assisted in developing the internal agent management system, and underwent the coding, testing, and optimization of the exception handling module
- Developed functional modules of the system according to project requirements and design schemes, including embedded software development, driver development, bare metal program development, and embedded operating system application development
- Participated in maintaining and upgrading internal software systems, resolving technical issues including automated drift scripting for JumpServer and startup self-check script development

IT Development Assistant Engineer, SmooRe (Shenzhen) International Holdings Limited

07/2024-08/2024

- Employed Vue 3 (Composition API) and TypeScript to build the user interface, and optimized the component structure of the page to enhance page loading speed and response efficiency
- Constructed the backend logic to build a Spring Boot microservice architecture using Java, and assisted in designing and implementing API interfaces to support data communication between the front end and the back end
- Integrated databases MySQL into the data storage and management, to promote dynamic pricing, and improve data processing efficiency

PROJECT EXPERIENCES

The Setup and Operation of the Backend Servers and Databases

Programmer, 02/2025-05/2025

Personal Engagement:

Mastered the technical framework from Java Web to Spring Boot, and built a medium-sized backend service, adopting the classic three-tier layered architecture (presentation layer, business layer, data layer), achieving separation of concerns: the presentation layer handles front-end interactions, the business layer encapsulates core logic, and the data layer manages database operations and effectively enhancing the maintainability and scalability of the code

the Setup and Operation of the Backend Server for the Online Real-Time Weather Query Website

Programmer, 09/2024-01/2025

Personal Engagement:

Undertook the backend architecture design and core code implementation of the project, and led the completion of the early system modeling (including class diagrams, activity diagrams, etc.), successfully deploying the service to the online environment, and achieving a full-process closed loop from design, development to operation and maintenance

The Front-End Design of the Online Real-Time Bus Query Website

Programmer, 10/2024-12/2024

Personal Engagement:

Developed the login/search module (using HTML/CSS/JS/Vue), studied the interaction mechanism between the backend API and the database, and participated in UML diagramming, achieving a leap from functional development to system understanding, and possessing the ability to participate throughout the entire process

ADDITIONAL

- Languages: IELTS 7.0 (08/2025), Mandarin (Proficient)